

TELCO CUSTOMER CHURN ANALYSIS

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PROBLEM STATEMENT

Customer churn refers to customers discontinuing telecom services. Churn directly affects recurring revenue and increases acquisition costs. This project analyzes customer churn behavior to identify key drivers and high-risk customer groups, enabling actionable retention strategies.

DATASET DESCRIPTION

Dataset Name: Telco Customer Churn Dataset

Dataset contains:

- 7043 customer records
- 21 columns

Main variables:

Demographic Data

- Gender
- SeniorCitizen
- Partner
- Dependents

Customer Loyalty

- Tenure

Service Information

- InternetService
- TechSupport
- OnlineSecurity
- StreamingTV
- StreamingMovies
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Billing Information

- Contract
- PaymentMethod
- MonthlyCharges
- TotalCharges

Target Variable

- Churn

APPROACH / METHODOLOGY

Data Cleaning

Performed:

- Checked null values
- Converted TotalCharges to numeric datatype
- Verified dataset consistency

Customer Segmentation

Created:

- High Risk Customers
 - Loyal Customers
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KPI Analysis

Calculated:

- Churn Rate
- ARPU
- Revenue Lost
- Average Churn Tenure

ANALYSIS / MODEL

- **Churn Distribution Analysis**

The observed churn rate indicates a customer retention challenge.

- **Contract Type Analysis**

Month-to-month customers show the highest churn tendency.

- **Monthly Charges Analysis**

Higher monthly charges correlate with higher churn.

- **Payment Method Analysis**

Electronic check users show increased churn behavior.

- **Tenure Analysis**

Short-tenure customers are more likely to churn.

KEY INSIGHTS

- **Insight 1**

Month-to-month contracts are the strongest churn driver.

- **Insight 2**

High monthly charges increase churn probability.

- **Insight 3**

Electronic check customers are high-risk.

- **Insight 4**

Customers with shorter tenure churn more frequently.

- **Insight 5**

Lack of tech support and online security strongly correlates with churn.

CONCLUSION

Customer churn is primarily influenced by pricing, contract flexibility, service engagement, and customer tenure. Early intervention and targeted retention programs can significantly reduce churn risk.

RECOMMENDATIONS

- **Convert Monthly Customers to Annual Plans**

Offer discounts for longer contracts.

- **Improve First-Year Onboarding**

Provide onboarding assistance.

- **Encourage Auto-Payment Enrollment**

Reduce churn risk.

- **Bundle Security + Tech Support**

Increase service stickiness.

- **Monitor High-Risk Segments**

Use predictive alerts for proactive retention

VISUALIZATIONS

Dashboard Page 1

Executive Summary

Contains:

- Total Customers KPI
- Churn Rate KPI
- ARPU KPI
- High Risk Customers KPI
- Churn Pie Chart
- Contract vs Churn
- Payment Method Analysis

Dashboard Page 2

Risk Analysis

Contains:

- High RiskCustomers KPI
- Loyal Customers KPI
- Avg Churn Tenure KPI
- Revenue Lost KPI
- Internet Service vs Churn
- Tech Support vs Churn
- Online Security vs Churn
- Tenure Group vs Churn

- Revenue Segment vs Churn
- Slicers